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The State of User-Centered Design in the Space Domain through a Literature Scan of the Past IAF Publications

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Abstract

For decades, both scholars and the industry have investigated and recognized the role that User-Centered Design (UCD) plays in a new product, service, and value creation, in both physical and digital domains. There are multiple ways this approach is defined in the literature, including Human-Centered Design (HCD) and People-Centered (PCD) Design. However, in its essence, UCD is an iterative process in which designers and creators focus on the users and their needs in each phase of the creation process.

Historically, within the technology-driven space industry, this approach has been relatively rare. If applied, it focused on the human factors and interior design in relation to human spaceflight. This, however, has been changing due to space industry transformation towards more commercial business models and interdisciplinary approach.

This paper demonstrates the changing landscape of how the user- and human-centered methodologies were evolving within the space industry through a literature scan of the past IAF papers in the last 75 years. By analyzing the depth and frequency of the publications related to the topic, the authors contribute to the definition of what UCD means in the context of the space domain, and how its understanding has been changing over time. For the purpose of this paper, the authors developed a specific taxonomy system to cover different types of UCD applied to space, both in relation to space exploration and terrestrial applications. That covers the areas related to product and industrial design, architecture, service design, strategy, human to machine interface design (UX/UI), communication design, design research and theory, fashion design, future foresight and speculative design.

The thematic literature review uses both quantitative and qualitative approaches, performed in two phases: firstly, by automatically filtering the abstracts within the online IAF database using keywords based on a taxonomy in a semi-systemic way. In the second phase, a more in-depth analysis is performed on a selection of papers and a mapping of an evolution of each area is provided.

As of early 2024, there are 15,522 papers using the word “design” (31%), out of which 2,308 papers mention “human” and 692 discuss the “user”. By understanding the context in which these words are used, this paper contributes to our understanding of the trend underpinning the transformation of the space industry, as well as understanding the research gaps of using UCD in both upstream and downstream space applications.

Keywords: user-centered, human-centred, design, literature scan, discipline, keywords

Acronyms/Abbreviations

Comma-Separated Values (CSV), Hypertext Markup Language (HTML), Hypertext Transfer Protocol (HTTP), Human-centered design (HCD), International Astronautical Congress (IAC), International Astronautical Federation (IAF), International Council of Design (ICoD), International Council of Graphic Design Associations (Icograda), International Council of Societies of Industrial Designers (Icsid), National Aeronautics and Space Administration (NASA), User-Centered Design (UCD), World Design Organisation (WDO), User Experience (UX), User Interface (UI)

1. Introduction

The objective of this paper is to explore the evolution of User-Centered Design (UCD) methodologies within the space industry by conducting a literature scan of the past International Astronautical Federation (IAF) publications. UCD, also referred to as Human-Centered or People-Centered Design, emphasizing an iterative approach where the needs of users are central throughout the design process [1]. While UCD has long been recognized as critical in physical and digital product creation, its application in the technology-driven space domain has historically been limited, largely focusing on human factors related to spaceflight. However, with the industry's shift towards commercial business models and a more interdisciplinary approach, UCD is gaining relevance across various domains.

This paper is the first one to examine how UCD methodologies have been discussed and implemented in the space industry over the past 75 years by analyzing the depth and frequency of related topics in IAF publications. A specific taxonomy system was developed to categorize different UCD applications, ranging from product and industrial design to future design and human-machine interfaces (UX/UI). The analysis is conducted through a two-phase review process: an initial quantitative filtering of abstracts followed by a qualitative deep dive into selected papers, mapping the evolution of UCD and the disciplines using this approach in space-related contexts.

In doing so, this paper aims to contribute to the understanding of UCD’s role in the changing landscape of space exploration and terrestrial applications, identifying trends and research gaps that can inform future innovations in both upstream and downstream space sectors.

2. Historical context

The space industry, a realm of engineers, has undergone a remarkable transformation over the past six decades. From Moon exploration to voyages to the outer edges of our solar system, engineers have pushed the boundaries of possibility. Today, the focus has shifted towards returning to the Moon and setting foot on Mars. In this ever-evolving landscape, the role of design in the space industry has grown increasingly significant. Design is no longer confined to our living rooms, smartphones, or streets—it has found its place in the vast expanse of space itself.

Historically, the integration of design in the space industry was limited. Two notable exceptions were Galina Balashova and Raymond Loewy. Balashova, an architect working for the Soviet space program from 1963 to 1991, left an indelible mark on space design. Her interior design projects for spacecraft like Soyuz, Saliout, and Mir showcased her innovative approach. Balashova's work prioritised user-centred design, aiming to create comfortable and functional living spaces for cosmonauts. She introduced elements like earthy colour palettes to aid spatial orientation and developed an ingenious Velcro fastening system to secure objects in the weightless environment [2].

Similarly, Raymond Loewy, a renowned industrial designer, made significant contributions to the American space program. In 1967, he collaborated with NASA (National Aeronautics and Space Administration) on the Skylab project, where he focused on enhancing human comfort in confined spaces. Loewy's emphasis on living conditions inside the spaceship led him to consider factors such as meals, privacy, relaxation, and morale. He challenged traditional thinking and pushed NASA to prioritise the astronauts' well-being. Notably, Loewy's proposal for a porthole, initially met with scepticism, eventually became a symbolic connection to Earth for astronauts and a testament to the importance of human-centric design in space [3].

Balashova and Loewy made pioneering strides in space design. However, their work exists within a broader trend in history of human-centered design, rooted in the mid-20th century which emphasizes the importance of empathy, usability, and iterative problem-solving. Pioneers such as Charles and Ray Eames, along with Herbert Simon, helped shape the early development of human-centered design, emphasizing the need to design for people first. It got popularized further through design thinking frameworks in the 2000s and 2010s which saw a significant growth of interest in applying design thinking across a range of diverse applications – for example, as a catalyst for gaining competitive advantage within business. [4] This approach has since expanded into areas like UX (User Experience) and UI (User Interface) design, which now play critical roles in developing intuitive and seamless experiences and interfaces for complex systems in all major industries.

As the space industry has evolved, so has the integration of design, particularly with the emergence of 'new space' initiatives with more commercial approach. This renewed focus on human-centered design is driving advancements across various disciplines, including strategic design, which aligns space innovations with long-term sustainability goals, and service design, which seeks to create user-centered services that benefit both space missions and life on Earth.

The impact of design in the space industry extends far beyond aesthetics; it has the power to transform the human experience in space. Whether through Galina Balashova's focus on human comfort and functionality or Raymond Loewy's advocacy for astronauts' well-being, design has brought a fresh perspective to space exploration. As the industry continues to evolve, design will play an increasingly pivotal role in shaping the future of space exploration and commercial services, driving sustainability and enhancing life on Earth by applying space innovations to address global challenges like climate monitoring, resource management, and disaster response.

By annually organising the International Astronautical Congress (IAC) and other conferences, the International Astronautical Federation (IAF) plays a significant role in shaping the future of the space sector globally. In order to understand the role of design in the space industry, it is therefore important to understand how much design has been present at those events and published as proceedings available in the IAF Digital Library (https://dl.iafastro.directory/).

3. Methodology

For the purpose of this paper, the authors developed a specific taxonomy system to cover different types of UCD applied to space, both in relation to space exploration and terrestrial applications. That covers the areas related to Industrial Design, Communication Design, Interior Architecture and Interior Design, Service Design, Strategic Design, User Interface Design, User Experience Design, User Research, Fashion Design, Future Foresight, and Engineering Design. The definitions of each discipline can be found in Appendix A.

The thematic literature review uses both quantitative and qualitative approaches, performed in two main phases: firstly, by automatically filtering the abstracts within the online IAF database using keywords based on a taxonomy in a semi-systematic way. In the second phase, a more in-depth analysis is performed on a selection of papers and a mapping of an evolution of each area is provided.

Firstly, a broad search using the umbrella term "design" was conducted to filter potentially relevant results across all the papers in the IAF database. Comparative analysis between "design" and more specific terms (e.g., "Communication Design") confirmed the broader search's effectiveness, with a 10:1 result ratio.

Secondly, a Python-based script, specifically designed for this exercise, automated the data extraction process, using requests for HTTP (Hypertext Transfer Protocol) calls, BeautifulSoup for HTML (Hypertext Markup Language) parsing, and pandas for structuring the metadata (article titles, authors, publication years, abstracts, and Paper IDs). The extracted data was organized into a comprehensive data frame as the master dataset. To ensure comprehensive results, the extraction iterated through potential result pages up to a high page limit (e.g., 500,000), circumventing limitations of only retrieving data from the first few pages.

Later, the authors of the paper, design practitioners and academics from different design disciplines selected keywords associated with each design discipline (e.g., "ergonomy" and "prototype" for Industrial Design; "user journey" and "empathy" for UX Design etc.) that formed the basis for the analysis. Some general keywords describing the user-centred methodology were also selected which included: human, user, centred, creative, product, holistic, inclusive, collaborative, universal, iterative, co-design, co-creation, usability. At this stage, all articles containing the term "design" needed to be cross-referenced with these general and discipline-specific keywords to categorize them into relevant fields.

As a first approach to that exercise, a second Python script was created. It filtered the dataset based on discipline-specific terms within the abstract and metadata, creating separate files for each discipline. For example, "ergonomy" and "prototype" categorized articles under Industrial Design, while "service blueprint" and "user journey" identified Service Design-related works. The final output included individual CSV (Comma-Separated Values) files for each discipline, summarizing articles that matched the keyword criteria. Each CSV contained detailed metadata reflecting contributions to each design field within the IAF Digital Library, supporting a broader understanding of the evolution and impact of these disciplines within the space sector. The code and the created CSV files can be accessed here:

https://drive.google.com/drive/folders/1QvfMs8bMF9TOyejBXkmE-Ce_M6PWqFgp?usp=drive_link.

The second approach aimed for more transparency of the process and keywords validation which proved to be key to the accuracy of the final results. Based on the created CSV dataset of all the articles containing the term "design", a dedicated spreadsheet was created for each design discipline, incorporating corresponding filters to enhance transparency and ensure the data's accessibility for manual validation. The process of manual validation involved the random selection of 50 papers from the database, which were then reviewed by design experts from the Designers in Space Community to assess their relevance to each respective discipline. These spreadsheets, designed to facilitate human-readable data analysis, are available for access through the following link:

https://drive.google.com/drive/folders/1ePsX6vqnAlX2OzUfOcoX3dpmpopAxREz.

4. Theory and taxonomy considerations

Design in aerospace is multifaceted, encompassing both technical and human-centred elements. HCD focuses not only on users and their needs but also includes commercialization, strategic models, and human-focused research. This expanded definition illustrates the blurred boundaries between not only specific design disciplines but to what extent do HCD methodologies need to be present in projects to be classified as an example of design thinking? For instance, while engineering design emphasises functional technical aspects, design engineering often integrates creativity and user focus into the process. Similarly, systems engineering may overlap with design thinking, both often considering entire project systems, yet their approaches differ in scope and application.

Several established sources list various design disciplines along with their associated domains and purposes. Attempts to map design disciplines are present in the literature. Edeholt and Joseph (2022), for example, explore the role of design disciplines in addressing climate change [5]. Similarly, Sanders (2008) created a Venn diagram, (see Fig.1) to map the intersections of design practice and research, highlighting the collaborative and evolving nature of design [6]. Another, more recent approach, was proposed by the Designers in Space Community in the form of Design Spectrum (2022) to be used as a practical guide to matching design methodologies to a wide range of space sector business challenges (Appendix C) [7].

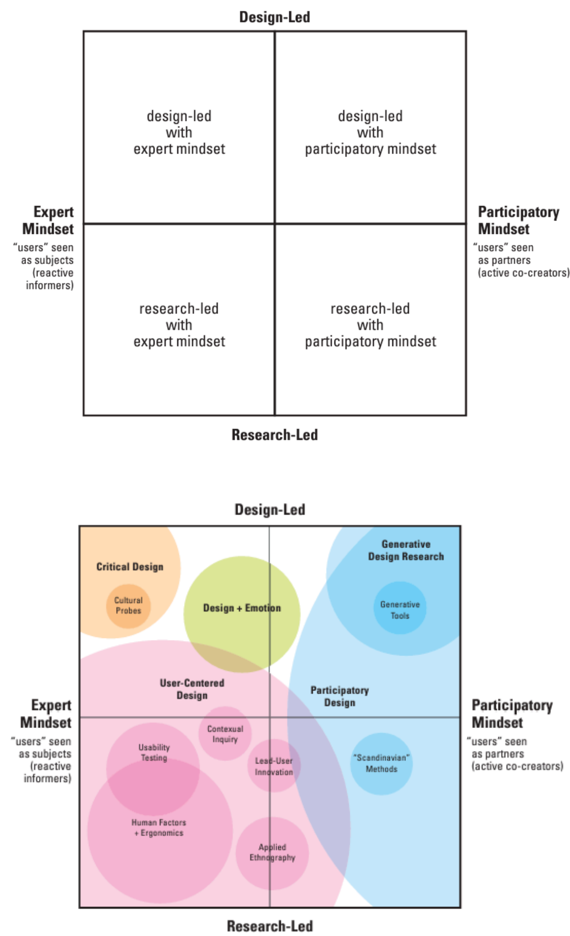

Fig.1: Sanders (2008) diagrams highlighting the intersection of various design practice and research

The globalisation of design and advancements in digital technologies have contributed to the obsolescence of traditional disciplinary frameworks. As Rodgers and Bremner (2017) argue in their manifesto for the future of design, "design now needs to disrupt, contest, invent, direct, coordinate, respond, provoke and project" using methods that are 'issue- or project-based' rather than 'discipline-based' [8]. This idea aligns with the interdisciplinary and adaptive nature of design in aerospace, where the boundaries between fields are increasingly blurred. Service design principles have been applied to space architecture and apparel design. User research, a discipline that is inherently multidisciplinary, is another area that transcends project-specific boundaries.

For this study, design disciplines were included if they were defined by an official organisation or governmental body. One such example is the World Design Organization (WDO), which defines industrial design as "a strategic problem-solving process that drives innovation, builds business success, and leads to a better quality of life through innovative products, systems, services, and experiences" [9]. Keywords associated with these disciplines were identified through secondary and primary research involving human-centered design experts. These experts categorised the subjects based on their formal education and design networks, ensuring that the taxonomy reflected the breadth of modern design disciplines. A general list of overarching design terms were also listed at the bottom in Appendix B.

The IAC database was then automatically filtered procuring that all abstracts included at least one keyword from their design discipline (full list in Appendix A) and at least one keyword describing the general UCD approach (see the list in Appendix B). Discipline key words were then revised to ensure generated lists were reliably related to that corresponding discipline. Key words that lacked a user-centred focus or provided no functional value—such as patches, logos, or general art—were excluded to maintain clarity and relevance to aerospace design.

Certain fields were deliberately excluded from the taxonomy due to their lack of relevance to user-centred design. These include Engineering Design, System Engineering, Marketing and Material Design. We also excluded Architecture from this exercise as it already has a strong presence within the sector and it could alter the results. Although all design disciplines can impact both users, systems, and technology (see Fig.2), this paper included disciplines where their primary initiatives are driven by enhancing human well-being through prioritising user focused strategies.

Recognizing that design is inherently contextual, varying significantly based on the domain in which it is applied, it is noted that this taxonomy serves as a pilot study and not exhaustive in its definition of all design disciplines.

For a detailed breakdown of design disciplines included in this study, see table in the Appendix A.

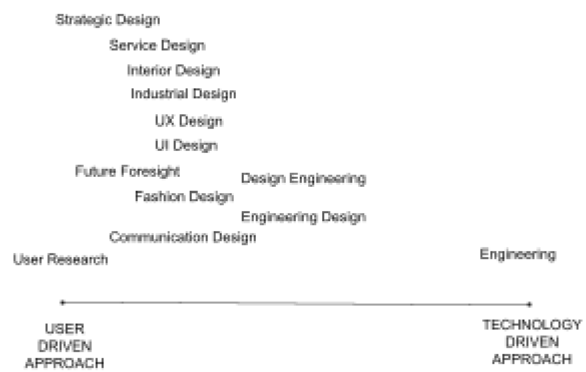

Fig.2. Comparison of various design disciplines implementation of user vs technology driven approaches

5. Results

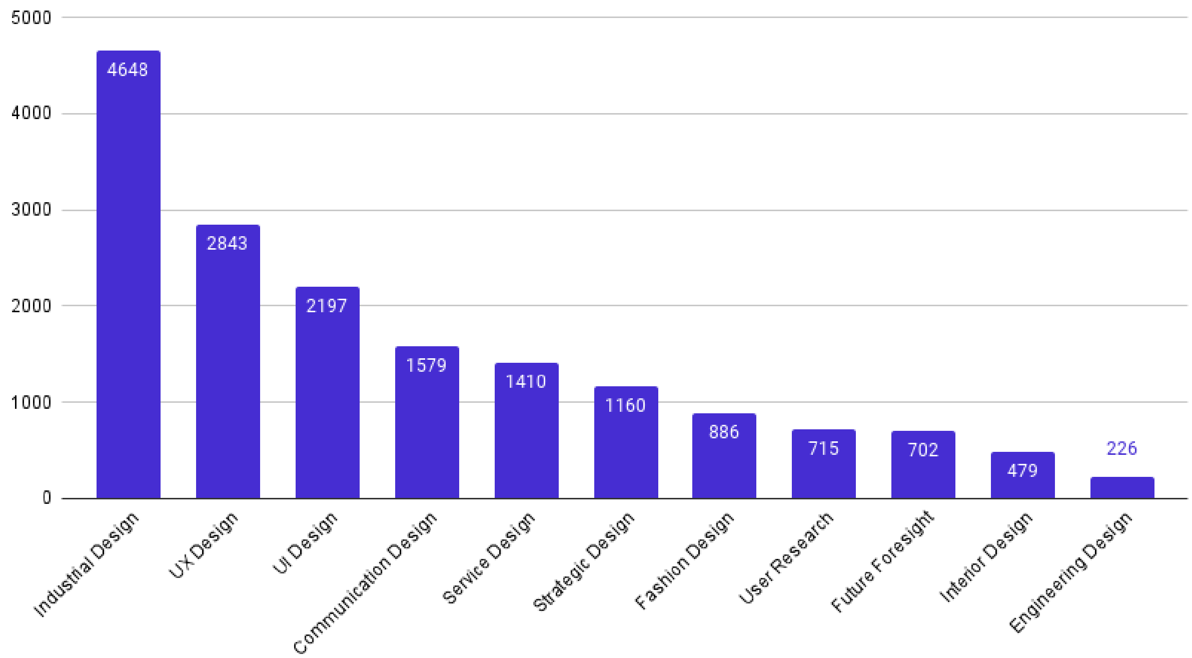

Fig. 3. Total number of IAF publications per discipline.

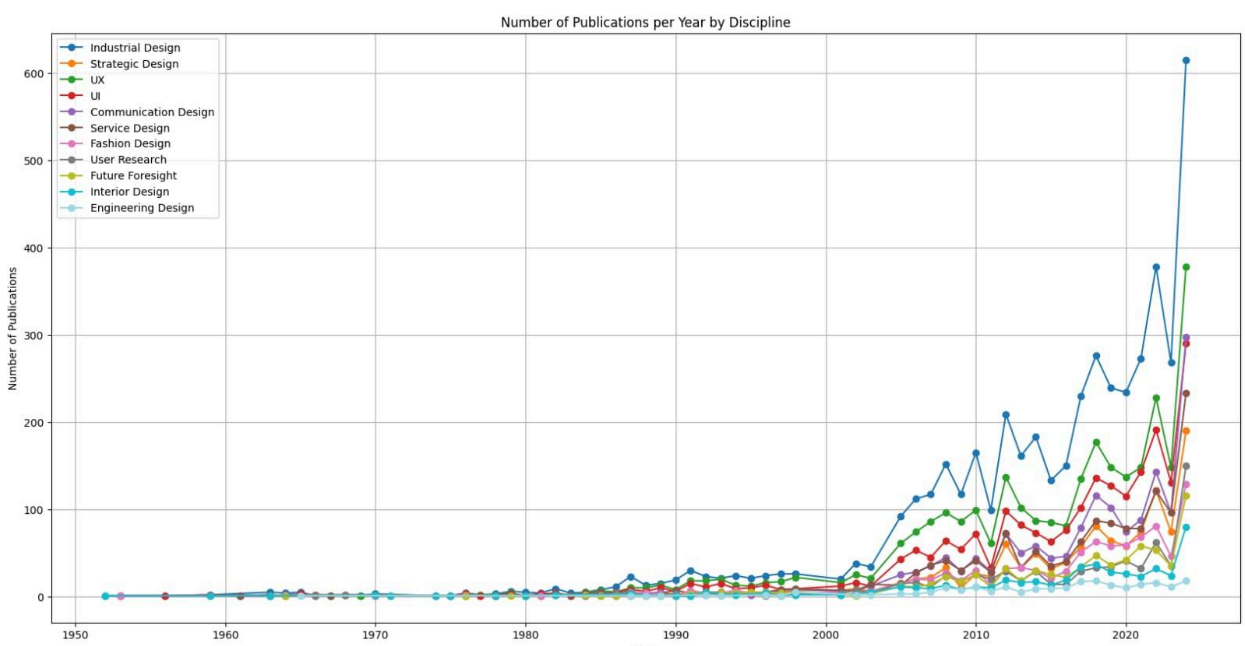

Fig.4. Results of the quantitative analysis: number of IAF publications per year and per discipline.

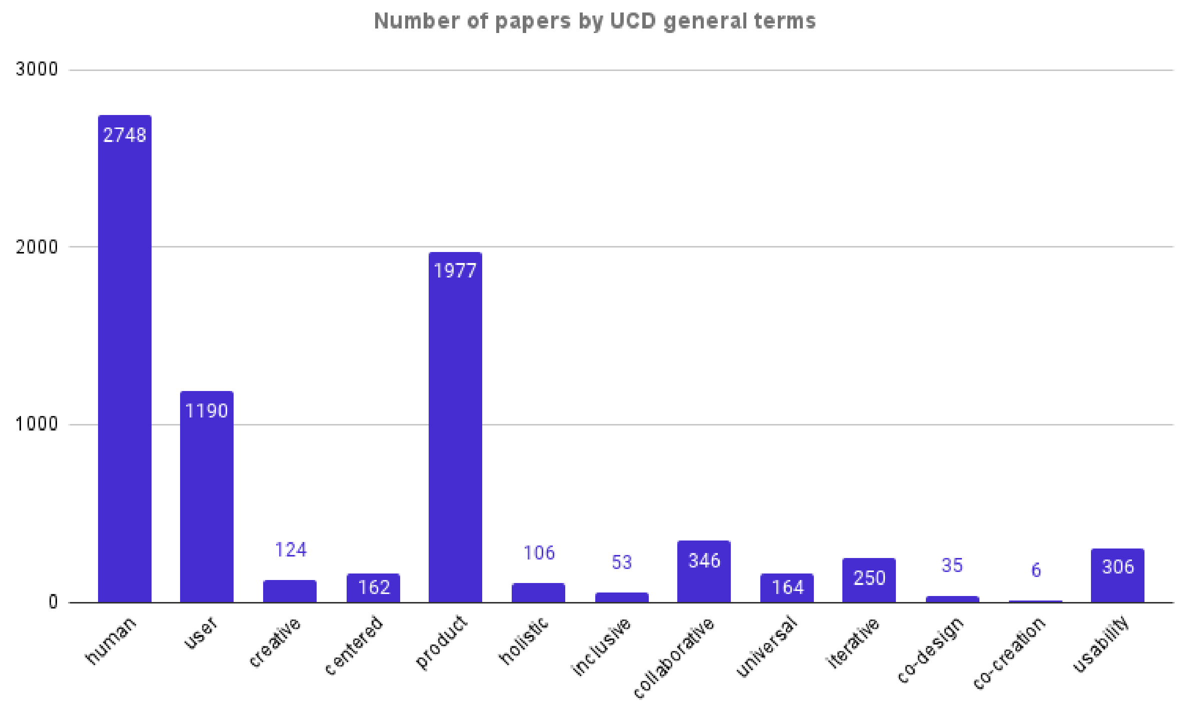

Fig.5. Number of publications by UCD general terms.

Table 1. Qualitative results of the filtering exercise.

Discipline	Total nr of publications	%	Filter accuracy	Papers not detected
Industrial Design	4648	27.02%	50%	6%
UX Design	2843	16.53%	66%	10%
UI Design	2197	12.77%	75%	10%
Communication Design	1579	9.18%	46%	12%
Service Design	1410	8.20%	80%	6%
Strategic Design	1160	6.74%	75%	18%
Fashion Design	886	5.15%	9%	18%
User Research	715	4.16%	50%	14%
Future Foresight	702	4.08%	50%	8%
Interior Design	479	2.78%	86%	18%
Engineering Design	226	1.31%	80%	8%
Average	NA	NA	**61%**	**12%**

The final results are presented in a form of the graphs showing the total numbers of IAF publications per discipline (Fig.3) and the distribution of IAF papers related to the analysed disciplines per year (Fig.4), showing the tendency over time.

Industrial Design has been the most popular discipline, with a total of 4,684 references found out of the 17,200 papers analyzed (27.02%). It is followed by User Experience Design (UX), with 2,843 publications (16.63%). Third place is held by User Interface Design (UI) with 2,197 results (12.77%).

The number of design-related papers began to increase just before the 1990s and entered an exponential growth trajectory in the early 2000s, which corresponds to the popularization of the design thinking framework.

Industrial Design continued to be the most cited discipline, which can be explained by its historical relationship with architecture, already a well-established discipline in the space industry.

UX (User Experience) and UI (User Interface) design remained the next most popular topics, likely due to the increasing digitalization of the space industry and the need for improved, user-friendly human-to-machine interfaces.

As explained in Section 3: Methodology, after automatic filtering by keywords, a manual validation based on a sample of 50 randomly selected papers determined the accuracy of the method and the margin of error. The results of this exercise are presented in Table 1. The average accuracy of the exercise (calculated by dividing the number of manually validated papers by the total number of papers identified by the filter) is 61%, meaning that 61% of the automatically filtered papers were relevant to the discipline in question. However, some relevant papers were missed by the automatic search, representing an average of 12% of the sample.

It is worth noting that there are significant differences in the accuracy of search results across disciplines (e.g., Communication Design with 46% accuracy vs. Service Design with 80%). These differences depend on:

- The quality of the keywords: Well-defined, descriptive keywords that are relevant and unique to the discipline, as well as consideration of different spelling variations and forms of words (e.g., “prototype” vs. “prototyping”), can return different results.
- The number of keywords: Interestingly, disciplines with a larger number of keywords returned more inaccurate results than those with fewer, more precisely defined keywords.
- Approach: The analysis used a unique approach, combining at least one UCD general keyword with at least one discipline-specific keyword. However, this approach may not be appropriate for disciplines such as fashion design, where another strategy - such as using at least two discipline-specific keywords - might have yielded better results.

Fig.6. Design Disciplines Wordcloud in terms of the number of articles found.

Fig.7. General Design Methodology terms

6. Discussion

UCD, once centred on optimising user-environment interactions, is evolving into a broader framework within the space industry. Originally focused on physical product design and human factors in spaceflight, UCD now plays a critical role across aerospace, driven by an interdisciplinary approach. This evolution reflects a shift in the space sector, particularly as commercial models gain traction and the industry's focus expands to include more diverse stakeholders and applications.

By conducting a comprehensive review of International Astronautical Federation (IAF) publications, we found that UCD methodologies are increasingly being discussed across a variety of design-related disciplines. This trend suggests that design in aerospace is becoming more holistic, bridging the gaps between areas such as industrial design, user experience (UX), architecture, and engineering. Rather than being confined to isolated fields, the application of design thinking now encompasses a broader range of disciplines, making UCD more about a mindset that integrates user needs and systems thinking at every level of space exploration and technology development.

Although our results show evidence of a strong upwards trend in the total number of design-related keywords used in IAF publications over time, it should be noted that these numbers are absolute and not relative. A search on IAF's digital database for all publications from the year 2022 provided 3149 results. For 2002, 851 results. For 1982, 281 results and for 1962, 63 results. Further study would be required to identify whether the amount of publications containing design-related keywords might just have increased together with the overall increase of IAF publications.

A major assumption on which our conclusion relies, which has not been tested in this study, is that publications at IAF conferences reflect the reality of the wider aerospace industry. Although IAF conferences are known for their international contributions from all over the industry, it is to be expected that a significant proportion of what happens within the industry is not reflected in public conferences due to confidentiality restrictions. It is also unclear whether the scope of IAF conferences might have changed over the last seven decades, which could influence the accuracy of the conclusions we have drawn regarding trends over time.

Validating our findings posed challenges due to the complexity and subjectivity of design disciplines. Unlike engineering, where results are quantitative, UCD relies on qualitative insights, making it harder to measure and validate. Our two-phase process - combining quantitative filtering and qualitative analysis - highlighted the difficulty of drawing clear boundaries between disciplines, as many papers overlapped across areas like industrial design, UX, and architecture.

Architecture, in particular, plays a prominent role in space design, often overshadowing UCD methodologies when it comes to spacecraft and habitat design. Our analysis showed that architecture frequently appeared in search results, even when the focus was intended to be on more user-centered approaches. This reflects the dominant role architecture has historically played in the design of space environments and indicates a need for clearer distinctions when evaluating the contribution of UCD within these contexts.

Finally, addressing the subjective nature of design validation and resolving the overlaps between disciplines will be necessary to understand better how UCD is evolving in this sector. More transparency in the keyword selection process and additional manual validation from diverse design experts will help to refine the process, ensuring more accurate representations of UCD methodologies in future research.

Keywords were crucial in shaping our findings, but their varying meanings within aerospace introduced challenges. For instance, “interface” could refer to human-machine interfaces (UX/UI) or mechanical interfaces between spacecraft components, introducing ambiguity in keyword-based categorization.

Ambiguities in terms like “persona” or “fashion” also led to manual adjustments to avoid false positives. The iterative process of keyword validation, supported by expert review from the Designers in Space Community, helped to mitigate these challenges but also highlighted the subjective nature of this categorization effort.

The coincidence approach, which aimed to identify overlaps between disciplines, proved effective in highlighting the interdisciplinary nature of UCD in aerospace. While it successfully captured cross-disciplinary themes, the approach struggled to categorise more niche disciplines, such as fashion design or future foresight, which are less frequently discussed in aerospace literature. These fields are harder to map using the current taxonomy system, as their terms often have multiple meanings across industries. For instance, “pattern” or “tailoring” might be used in contexts unrelated to design, such as describing trends or styles of operation.

Moreover, the language used in the earlier editions of conferences like the International Astronautical Congress (IAC) added further complexity. Historically, papers were published in French and German, which may have excluded relevant work in those languages. The shift to predominantly English-language publications narrows the field of review, potentially limiting the breadth of applicable research in the field.

As the aerospace industry continues to embrace UCD principles, a few key trends emerge. First, the interdisciplinary merging of design disciplines is likely to continue, especially as the space sector moves toward more user-driven, commercially viable solutions. Second, the need for refined keyword taxonomies and more accurate validation methods is critical for future research. Enhancing these tools will improve the precision and relevance of UCD studies in aerospace, ensuring that all applicable design disciplines are appropriately represented.

Finally, addressing the subjective nature of design validation and resolving the overlaps between disciplines will be necessary to understand better how UCD is evolving in this sector. More transparency in the keyword selection process and additional manual validation from diverse design experts will help to refine the process, ensuring more accurate representations of UCD methodologies in future research.

In conclusion, our results show that UCD keywords are increasingly used in IAF conference publications. This might reflect an overall industry trend in which design methodologies are more widely applied for aerospace applications. However, further studies would be necessary to gain more accurate statistical insight into the absolute versus relative increase of keywords compared to total publications, and to what extent publications at IAF conferences reflect a reality within the aerospace industry as a whole.

7. Conclusions and future work

This paper provides valuable insights into the evolving role of User-Centered Design within the space industry by analyzing past IAF publications. By examining the context in which key terms such as “design,” “human,” and “user” are used, we shed light on the trends driving the industry's transformation. This analysis also highlights the research gaps in applying UCD in both upstream and downstream space applications, offering a foundation for future exploration.

To our knowledge, this is the first quantitative study of its kind to investigate design disciplines within the space sector. The findings indicate a growing recognition of UCD principles, but further research is needed to deepen our understanding. Future studies could expand by analyzing a broader range of paper types based on symposium themes, such as downstream applications, human spaceflight, and architecture, or by using keywords more systematically. Additionally, examining the geographical distribution of authors could provide useful insights into regional trends. Finally, exploring keyword pairs and combinations, such as “design” and “human,” would offer a more nuanced understanding of how these concepts intersect in space-related research.

REFERENCE LIST

[1] International Council of Design. What is Design?, https://www.theicod.org/en/professional-design/what-is-design/what-is-design (accessed 07.07.23).
[2] P. Meuser, Galina Balashova: Architect of the Soviet Space Programme, Dom Publishers, 2015
[3] A.Schonberger, Raymond Loewy: Pioneer of American Industrial Design, Prestel, 1990
[4] T. Brown, Design Thinking. Harvard Business Review, June 2008, pp. 85–92.
[5] H. Edeholt, J. Joseph, Design disciplines in the age of climate change: Systemic views on current and potential roles, in Lockton, D., Lenzi, S., Hekkert, P., Oak, A., Sádaba, J., Lloyd, P. (eds.), *DRS2022:* Bilbao, 2022, 25 June - 3 July, Bilbao, Spain. https://doi.org/10.21606/drs.2022.365 (accessed 07.07.23).
[6] L. Sanders, An evolving map of design practice and design research, interactions, 15/6 (2008), 13–17.
[7] Community, Designers in Space. “The Space and Design Innovation Spectrum | by Designers in Space Community | Medium.” *Medium*, Medium, 9 Sept. 2022, https://medium.com/@designersinspacecommunity/the-space-and-design-innovation-spectrum-a0db5de96b0b.
[8] P. Rodgers, C. Bremner, The Concept of the Design Discipline, Dialectic, I (1, Winter 2017).
[9] “Definition of Industrial Design.” *WDO*, World Design Organization, https://wdo.org/about/definition/. Accessed 20 Sept. 2024.
[10] World Design Organization, Definition of industrial design, https://wdo.org/about/definition/ (accessed 07.07.23).
[11]] International Federation of Interior Architects/Designers. Advocacy Platforms, https://ifiworld.org/advocacy-platforms/ (accessed 07.07.23).
[12] P. Rodgers, C. Bremner, The Concept of the Design Discipline, Dialectic, I (1, Winter 2017).
[13] International Federation of Interior Architects/Designers, IFI Interiors Declaration (IFI ID), https://ifiworld.org/wp-content/uploads/2020/09/IFI-Interiors-Declaration_ENGLISH_rev-seal-2020.pdf (accessed 07.07.23).
[14] B. Mager, Service Design, in Design Dictionary, T. Marshall, M. Erlhoff., 2008, p. 354-357,

https://www.academia.edu/20380596/Service_Design_Definition?source=swp_share
[15] A.Simon, The Space and Design Innovation Spectrum, Designers in Space Community, Medium, 2022, https://medium.com/@designersinspacecommunity/the-space-and-design-innovation-spectrum-a0db5de96b0b
[16] "Recommended Readings." *WDO*, World Design Organization , https://wdo.org/resources/recommended-readings/. Accessed 20 Sept. 2024.
[17]"BEDA - The Bureau of European Design Associations." *BEDA - The Bureau of European Design Associations*, https://beda.org/. Accessed 20 Sept. 2024.

Annex A: Design definitions

*Discipline definitions have been based on references from a body or professional organisation expert within this discipline.
**Keywords: selection of descriptive concepts, tools and approaches associated with a given discipline that created the base for the database filter.

Design discipline	Organization	Definition*	Source	Keyword selection**
Industrial design	World Design Organisation (WDO), formerly known as the International Council of Societies of Industrial Designers (Icsid, 1957-2017) Industrial Designers Society of America (IDSA)	"Industrial Design as defined by the World Design Organisation, is a strategic problem-solving process that drives innovation, builds business success, and leads to a better quality of life through innovative products, systems, services, and experiences. Industrial Design bridges the gap between what is and what's possible. It is a trans-disciplinary profession that harnesses creativity to resolve problems and co-create solutions with the intent of making a product, system, service, experience or a business, better. At its heart, Industrial Design provides a more optimistic way of looking at the future by reframing problems as opportunities. It links innovation, technology, research, business, and customers to provide new value and competitive advantage across economic, social, and environmental spheres." "Industrial Design (ID) as defined by the Industrial Designers Society of America (IDSA), is the professional practice of designing products, devices, objects, and services used by millions of people around the world every day. Industrial designers typically focus on the physical appearance, functionality, and manufacturability of a product, though they are often involved in far more during a development cycle. All of this ultimately extends to the overall lasting value and experience a product or service provides for end-users.	https://wdo.org/about/definition/ https://www.idsa.org/about-idsa/advocacy/what-industrial-design/	brainstorm, product, materials, manufacturing, fabrication, costs, product life cycle, aesthetics, shape, form, texture, color, interaction, usability, modularity, form follows function, concept, (conceptual design), ergonomic, usability, functionality, prototype, prototyping, iterative design, trans-disciplinary, creative, competitive advantage, design process, (product development)

		Every object that you interact with on a daily basis in your home, office, school, or public setting is the result of a design process. During this process, myriad decisions are made by an industrial designer (and their team) that are aimed at improving your life through well-executed design.”		
Communication Design	British Academy of Graphic Design	*This category also includes visual design, creative communication, and graphic design.* Visual communication as defined by British Academy of Graphic Design is the use of images, typography, colours, and layout to convey messages and share information. It encompasses everything that defines graphic design as it is the means by which designers communicate with their intended audience. Colour, typography, images, shapes, lines, and textures are all important elements of visual communication with each aspect working in harmony to come together to tell a story. Some of the fundamental principles of visual communication include balance, contrast, hierarchy, and unity. Combined with a designer's vision, it is these principles that will help create a successful design that can stand the test of time. For other definitions of design see the American Institute of Graphic Arts (AIGA), and Design and Art Direction (D&AD)	https://www.graphicdesignacademy.uk/blog/how-to-use-visual-communication-in-graphic-design-and-why-it-matters	visualisation, visualization, typography, brand, illustration, information design, motion graphics, science communication, visual, digital, print, advert, marketing, content strategy, editorial design, graphic, multimedia, storytelling, engagement
Interior Design	International Federation of Interior Architects/Designers (IFI, 1963-present)	Defined by the International Federation of Interior Architects/Designers“[…] Interior Architecture / Design, which asserts that, by design, Interiors optimally support humanity in all of its functions and activities.” As per the International Interior Design Association, “Interior design is defined as the professional and	https://ifiworld.org/programs-events/interiors-declaration-adoptions/	structural, remodelling, structural regulations, architectural integration, acoustic design, building code compliance, spatial configuration, adaptive reuse, fire safety design, aesthetic,

	International Interior Design Association (IIDA, 1994*-present) * a merger of the Institute of Business Designers (IBD); the International Society of Interior Designers (ISID); and the Council of Federal Interior Designers (CFID)	comprehensive practice of creating an interior environment that addresses, protects, and responds to human need(s). It is the art, science, and business planning of a creative, technical, sustainable, and functional interior solution that corresponds to the architecture of a space, while incorporating process and strategy, a mandate for well-being, safety, and health, with informed decisions about style and aesthetics."	https://iida.org/about/what-is-interior-design	colour scheme, furniture layout, textiles, ornamentation, third space, third place, restoration
Service design	Prof. Birgit Mager of Köln International School of Design (KISD)	"Service design addresses the functionality and form of services from the perspective of clients. It aims to ensure that service interfaces are useful, usable, and desirable from the client's point of view and effective, efficient, and distinctive from the supplier's point of view."	https://www.academia.edu/20380596/Service_Design_Definition?auto=download	persona, service blueprint, user journey, touchpoint, storyboard, service design, framework
Strategic Design	Giulia Calabretta, Delft University, Gerda Gemser, RMIT University	*This category includes Business Design.* "The use of design principles and practices to guide strategy development and implementation toward innovative outcomes that benefit people and organisations alike."	http://strategicdesignbook.com/	business model canvas, strategy, business, prototyping, framework
User Interface design (UI)	human-computer interaction (HCI), HCI, pioneers such as Douglas Engelbart, Alan Kay, and Ivan Sutherland made significant contributions to the development of graphical user interfaces (GUIs) in the 1960s and 1970s.	User interface (UI) design is the process designers use to build interfaces in software or computerized devices, focusing on looks or style. Designers aim to create interfaces which users find easy to use and pleasurable. UI design refers to graphical user interfaces and other forms—e.g., voice-controlled interfaces. A user interface is the point of interaction between humans and computers. User interface *design* is the process of designing how these interfaces look and behave.	https://www.interaction-design.org/literature/topics/ui-design	digital, interaction, machine, interface, MMI, UI, flow, wireframe, prototype, information architecture, accessibility, usability, front-end

User Experience Design (UX)	International Council of Design (ICoD), formerly known as the International Council of Graphic Design Associations (Icograda, 1963-2014)	UX design considers the entire user experience from A to Z, always keeping the target users' needs, goals and pain-points in mind. The goal of UX design is to create products and experiences that are easy, efficient, enjoyable and rewarding for the end user. Alternative: (Communication) design as defined as the International Council of Design (ICoD): "Design is a discipline of study and practice focused on the interaction between a person — a 'user'— and the man-made environment, taking into account aesthetic, functional, contextual, cultural and societal considerations. As a formalised discipline, design is a modern construct."	https://www.uxdesigninstitute.com/blog/what-is-ui-design/ https://www.theicod.org/en/professional-design/what-is-design/what-is-design https://www.interaction-design.org/literature/topics/ux-design#what_is_user_experience_(ux)_design?-0	ux, experience, empathy, interaction, profile, journey, wireframe, needs, interface, digital, virtual, computer, website, front-end
User research	Interaction Design Foundation	User research is the methodic study of target users, including their needs and pain points, so designers have the sharpest possible insights to work with to make the best designs. User researchers use various methods to expose problems and find crucial information to use in the design process.	https://www.interaction-design.org/literature/topics/user-research	qualitative, quantitative research, evaluative research, user research, concept testing, usability testing, survey, user testing, interviewing, user interview, shadowing, ethnography, diary study, mental model, insight, behaviour
Fashion design	Council of Fashion Designers of America (CFDA), British Fashion Council (BFC), Camera Nazionale della Moda Italiana (CNMI), The Fédération de la Haute Couture et de la Mode, etc.	No official definition. "Fashion Designers plan, design and develop clothing, accessories, footwear or other items of personal apparel considering the form and construction of clothing, historical styles and contexts, contemporary and cultural trends, colour, fabric, and decoration, and the techniques and processes available for manufacture."	https://www.yourcareer.gov.au/occupations/232311/fashion-designer#overview	spacesuit design, fabric, textile, garment, pattern, cloth, protective gear, apparel, aesthetics, trend, collection, draping, footwear, tailoring, accessories, fashion, expression, style, pattern

Future foresight	The Association of Professional Futurists	Foresight, the competency and practice of futures studies, is the capacity to think systematically about the future to inform decision making today. Specifically, it focuses on investigating the drivers of change and exploring possible futures to inform planning and policymaking. Understanding, anticipating, and navigating different types of change is a core feature of foresight work, particularly disruptive events, future developments, risks, and opportunities. Foresight practitioners use systematic methods to understand change and develop alternative scenarios, such as environmental and horizon scanning, scenario development, futures wheel, trend analysis, and forecasting. Foresight does not seek to predict since none of the futures that come to pass are exactly as imagined. However, it does support better preparation for any future which may arise, as well as spur imagination and collective creativity.	APH Foresight Evaluation Task Force Report_ Building Field and Foresight Practitioner Evaluation Capacity_ 2022	scenario, trend, strategic, foresight, future, forecast, horizon scanning, speculative design, artefact, delphi, fiction
Design Engineering	Accreditation Board for Engineering and Technology, Inc. Dyson School of Design Engineering Imperial College London	*ED, as defined by ABET (Accreditation Board for Engineering and Technology, Inc.) "[...] is a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions.[...]" ** DE, as defined by Dyson School of Design Engineering Imperial College London "[...] is a highly creative discipline that draws on knowledge of manufacturing techniques, product development, technical design and rapid prototyping to bring new innovations to market, and to improve existing products and the processes used for making them."	https://www.abet.org/accreditation/accreditation-criteria/criteria-for-accrediting-engineering-programs-2022-2023/ https://www.imperial.ac.uk/design-engineering/about-us/what-is-design-engineering/	transdisciplinary, detailed design, integrated design, concept engineering, concurrent engineering, ethical engineering, double diamond, integrated design, integrated design, concurrent design

Annex B: User-Centred Approach Terms

The table below gathers definitions and related keywords to the user-centred approach and its synonyms.

*Definitions have been based on references from a body or professional organisation.
**Keywords: selection of descriptive concepts, tools and approaches associated with a given discipline that created the base for the database filter.

Approach name	Organization	Definition*	Source	Keywords selection
Human-centred design or User-Centred Design	ISO 9241-210:2019(en) Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems;	“Approach to interactive systems development that aims to make systems usable and useful by focusing on the users, their needs and requirements, and by applying human factors/ergonomics, and usability knowledge and techniques. This approach enhances effectiveness and efficiency, improves human well-being, user satisfaction, accessibility and sustainability; and counteracts possible adverse effects of use on human health, safety and performance.”	https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:v1:en	Human, user, centred, creative, product, holistic, inclusive, collaborative, universal, iterative, co-design, co-creation, usability
Universal design, Universal Design (UD) or Inclusive Design. UD is defined as “design for all people”	Ronald L. Mace, Center for Universal Design at North Carolina State University in the late 1980s	“The design of products and environments to be usable to the greatest extent possible by people of all ages and abilities/disabilities.”	The Universal Design File: Designing for People of All Ages and Abilities. Revised Edition. Story, Molly Follette; Mueller, James L.; Mace, Ronald L. https://eric.ed.gov/?id=ed460554	

Appendix C: DiSC Design Spectrum

Source: Designers in Space Community, on Medium (2022):
https://medium.com/@designersinspacecommunity/the-space-and-design-innovation-spectrum-a0db5de96b0b